

calculate and predict exactly how different parts of the building will perform. For example, HEVACOMP, a software package, simulates the energy performance of the building and enables architects to design the surfaces accordingly. It also calculates the heating and cooling load required per month, by taking into account external weather conditions.

PVSYST is another software package that helps architects design the size and type of the photovoltaic (PV) system required based on the orientation and slope of PV panels. This enables architects to visualise the building performance after basic elements of human well-being such as appropriate ventilation, lighting and thermal comfort have been incorporated.

Another very critical use of computer technology is in database management. Learning about building usage is based on a wealth of data that is continuously generated. Moreover, all data on sourcing of sustainable materials and technology of building materials need to be documented, shared and transferred to other projects for analysis. This helps in innovation of more energy-efficient buildings.

VR models for restoration of archaeological sites

The Virtual Reconstruction Project of the central temple at Sambor Prei Kuk (SPK) in Cambodia is the application of 21st-century technology to 7th-century cultural heritage. The immersive, interactive, Web-enabled, multi-user virtual environment (MUVE) has enabled patrons to narrate the story of SPK in a manner that can aid visitors relive the extraordinary cultural heritage

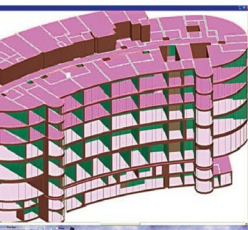


Fig. 3: Screenshot of Hevacomp Mechanical Designer V8i, the mechanical CAD software (Image courtesy: www.directindustry.com)

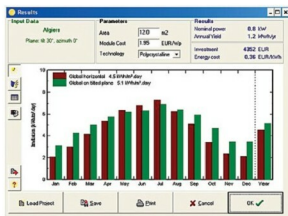


Fig. 4: Screenshot of PVsyst 6.0 beta in action (Image courtesy: pvsyst.software.informer.com)



Fig. 5: Digital reconstruction of the entry to SPK shrine (Image courtesy: ced.berkeley.edu)

as it was in 7th century AD.

MUVE is a new media vehicle that has the ability to communicate cultural heritage experience by integrating film-making, video games

and architectural design.

Today, a number of archaeological sites are being virtually reconstructed using graphical workstations. Measurements are sourced via a variety of 3D measurement techniques and input into virtual reality (VR) models.

Rebuilding the entire site manually using a modelling package is a tedious process. Hence, a more efficient way of doing it is to use images of the site as inputs. There are systems that can automatically reconstruct textured 3D models from image sequences of a site. This reconstruction is very much like virtualised reality. After the pictures have been taken, these are fed into a computer. Post digitising, all photographs are converted into a standard size by rescaling and/or cropping. Brightness and colour balance are made identical using image-editing programs such as Adobe Photoshop.

On UNIX OS, it is possible to give a command like ImageMagick to rescale or brighten an entire directory of images in one shot. The final process puts the photos together to form a VR model using an applet, which is generally free of cost and platform-agnostic.

VR models of buildings can also be used for analysing different interior design options, operations planning, evacuation simulations and the like. VR models is playing a significant role as an information hub. The end user can just click on a particular object such as a door and get all sorts of useful information like

measurements, colour, design and so on. Moreover, there is packaged software available to generate VR model based project engineering and design data.